Requirement Analysis

Date	2 March 2026
Team ID	
Project Name	Predictive Pulse: Harnessing Machine Learning Blood Pressure
Maximum Marks	4 Marks

✅ Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form
		Registration through Gmail
		Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email
		Confirmation via OTP
FR-3	Blood Pressure Data Management	Manual BP Data Entry
		Upload Historical BP Records
		Store Data in Database
FR-4	ML-Based Risk Prediction	Analyze BP Trends
		Predict Risk Level (Low/Medium/High)
		Generate Health Alerts
FR-5	Dashboard & Reports	Display Graphical BP Trends
		Show Risk Status
		Provide Basic Health Recommendations

🔒 Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a simple, user-friendly interface for easy data entry and monitoring.
NFR-2	Security	User data must be encrypted and protected with secure authentication mechanisms.
NFR-3	Reliability	The system should provide accurate predictions with minimal errors.
NFR-4	Performance	The system should generate predictions and load dashboard data within a few seconds.
NFR-5	Availability	The system should be accessible 24/7 with minimal downtime.
NFR-6	Scalability	The system should support increasing users and large health datasets without performance degradation.

📊 Analysis of Solution Requirements

The solution requirements for *Predictive Pulse – Harnessing Machine Learning for Blood Pressure Analysis* were identified through systematic analysis of user needs, healthcare challenges, and technical feasibility. The system is designed to ensure accurate prediction, secure data handling, and seamless user interaction.

From a **functional perspective**, the application must support secure user registration and authentication, efficient blood pressure data entry, reliable storage of health records, and real-time machine learning-based risk prediction. Additionally, the system must provide a clear dashboard interface displaying BP trends, risk levels, and health alerts to enable proactive monitoring.

From a **non-functional perspective**, the solution must ensure high usability through an intuitive interface, strong security through encrypted data handling and authentication mechanisms, and reliable

performance with minimal latency in prediction generation. Scalability and availability are also critical to support increasing users and continuous system access without degradation in performance.

Overall, the requirement analysis ensures that the system is user-centered, technically robust, secure, and capable of delivering accurate and timely health insights .

📌 Scope of the Solution

The scope of *Predictive Pulse – Harnessing Machine Learning for Blood Pressure Analysis* is to develop a web-based application that enables users to monitor and analyze their blood pressure readings using predictive analytics.

The system focuses on collecting user BP data, storing historical records, and applying machine learning algorithms to predict potential risk levels (Low, Medium, High). It includes user authentication, data visualization through dashboards, and alert mechanisms for abnormal readings.

The solution is designed for individual users and can be extended in the future to integrate wearable devices, doctor consultation modules, and cloud-based large-scale healthcare systems.

🎯 Objectives of the Solution

1. To develop a secure and user-friendly platform for blood pressure data management.
2. To implement a Machine Learning model for early risk prediction of hypertension.
3. To provide real-time insights and graphical trend analysis for proactive healthcare monitoring.
4. To ensure data security, reliability, and scalability of the system.
5. To promote preventive healthcare and reduce dependency on frequent hospital visits.